

Clean coal technology: gasification of South African coals

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ABSTRACT

Electricity demand in South Africa is increasing at a rate of 1000 MW per year. Whilst there is increasing pressure to adopt non-fossil fuel electricity generating technologies, the abundant reserves and low cost of coal make it the preferred energy source to meet increasing electricity demand for the foreseeable future. The challenge in the future is to enhance both the efficiency and environmental acceptability of coal use by adopting clean coal technologies (CCTs).

Integrated gasification combined cycle (IGCC) is a potential CCT that could be applied in South Africa to increase efficiency and reduce carbon dioxide emissions. IGCC also holds the advantage of reduced water consumption and the potential for co-production of liquid and gaseous fuels and chemicals.

Fine coal gasification is a key enabling technology for the implementation of IGCC plants. Fluidised bed gasification is being evaluated by the CSIR as a potential fine coal gasification process for incorporation into future IGCC plants.

A suite of four South African coals has been identified as being possible fuels for power stations which would operate for three or four decades, towards the middle of this century. This paper presents the results of coal characterisation, thermogravimetric analysis (TGA) and pilot plant gasification tests to ascertain the performance of the selected coals under fluidised bed gasification conditions.

Key words: coal, gasification, fluidised bed, characterisation, combined cycle, modelling

1 Introduction

South Africa's primary energy supply is made up of the following components: coal 74.1%, oil 12%, renewable energy (hydro, biomass, solar and wind) 7.4%, nuclear 4.2% and gas 2.3% [1]. Due to the high cost and decreasing reserves of oil and gas, its contribution to the energy mix is expected to decrease. Since South Africa is a water scarce country the contribution of renewable energy such as hydro and biomass is not expected to increase significantly. The use of solar and wind power is also currently limited by its high cost. Safety and cost are issues that inhibit the increased use of nuclear energy. Abundant and cheap coal reserves will therefore almost certainly remain our most important energy resource for the foreseeable future.

Based on scientific analysis it is generally accepted that a link exists between climate change and the use of fossil fuels such as coal. The development of CCTs has therefore received increased attention worldwide. CCTs are defined as "Technologies designed to enhance both the efficiency and the environmental acceptability of coal extraction, preparation and use" [1].

CCTs that are being developed for power generation include:

- Integrated gasification combined cycle technology (IGCC)
- Ultra supercritical pulverised coal combustion (SCPCC)
- Oxy-coal combustion
- Circulating fluidised bed combustion (CFBC)
- Post combustion capture

The CSIR has identified IGCC as a potential CCT that could be applied in South Africa to achieve near zero emissions of greenhouse gases, which is likely to be a requirement for electricity producers towards the middle of the 21st century.

2 Integrated Gasification Combined Cycle Technology

The flowsheets for conventional and IGCC power generation cycles are given in Figure 1. In a conventional cycle all the energy in the coal is used to generate steam which is then exhausted through a steam turbine to generate electricity. The exhaust steam has to be recondensed and recycled to the boiler. Due to large energy losses during condensation the overall efficiency (coal to electrical power) of a conventional power station is between 33% and 38% [2]. This can be raised to 45% - 47% by increasing the temperature and pressure of the steam (SCPCC) [3]. New high-strength materials are being developed to achieve this target.

In an IGCC power station a coal gasifier is incorporated into the flowsheet. During gasification coal is reacted with oxygen/air and steam to produce a combustible gas (syngas). This gas stream has a low volume compared to flue gas resulting from conventional coal combustion, and therefore gas clean-up systems can be reduced in size. The cleaned gas is combusted in a gas turbine that produces electrical power while heat is recovered from the turbine exhaust gas by means of a conventional steam cycle. This configuration (IGCC) produces higher efficiencies (45% - 55%) and lower emissions than conventional power stations [4]

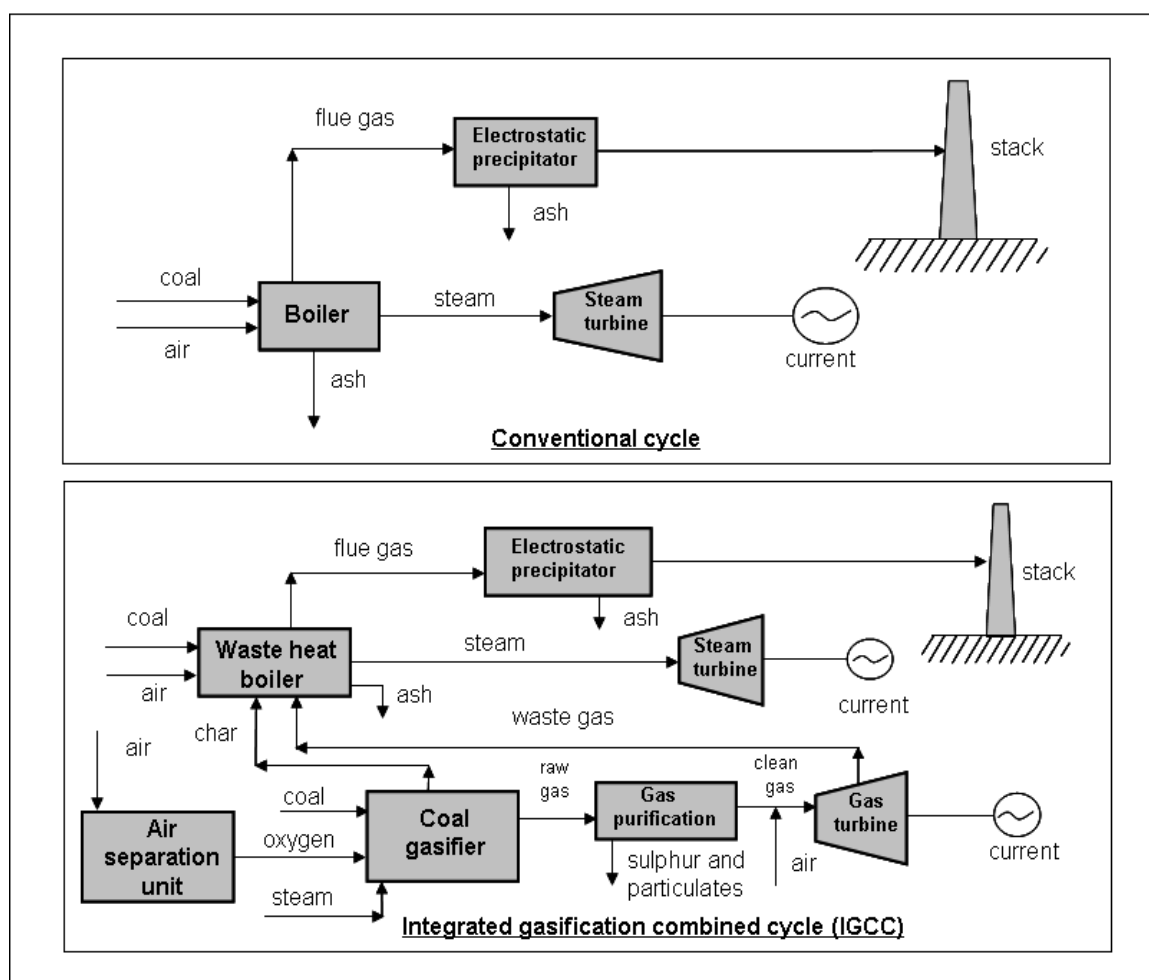


Figure 1. Conventional and IGCC power generation cycles.

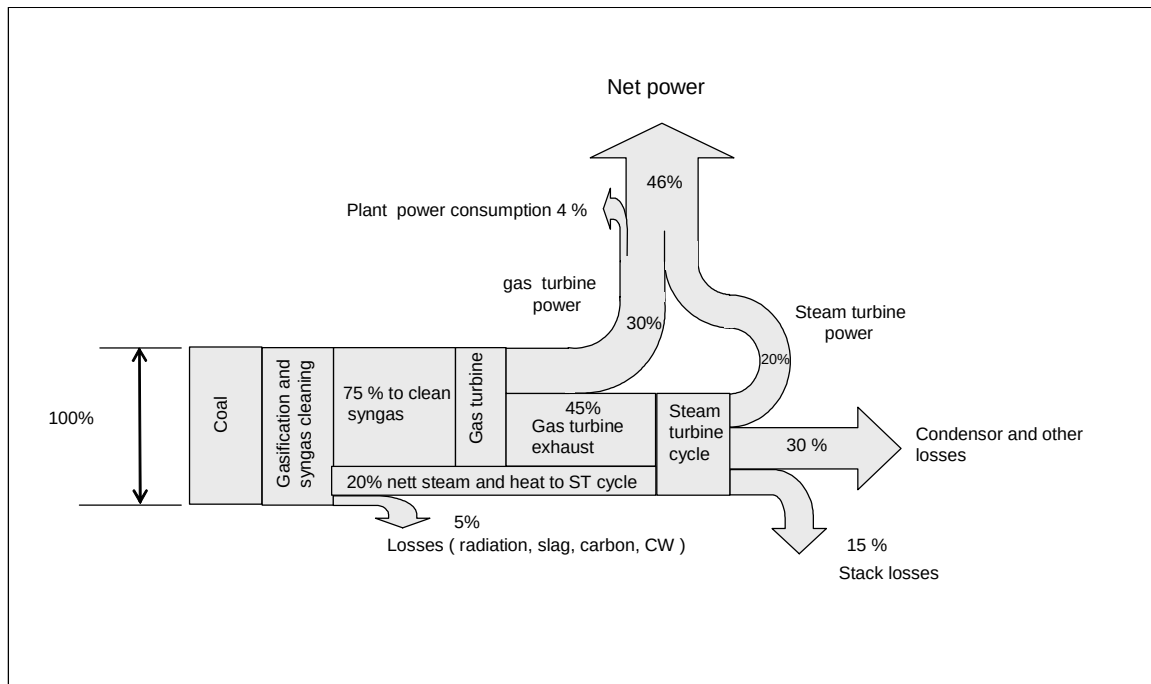


Figure 2. Energy flows in a IGCC power station.

The energy flows in an IGCC power station are given in Figure 2. The flow diagram shows that the coal to electric power efficiency of the power station is 46%.

3 Coal gasification

Coal gasification is a key enabling technology for IGCC plants. Coal gasification is not new to South Africa since Sasol operates 72 Lurgi gasifiers at

its synfuel plants in Secunda. The Lurgi gasifier, however, is more suited to synthetic fuel and chemicals production since it cannot utilize fine coal (< 12 mm) and byproducts which include tars, and oils are produced. For IGCC plants, fine coal gasification is the technology of choice [5]. The most well-known fine coal gasifiers are the entrained flow gasifier and the fluidised bed gasifier. These two types of gasifier are presented in Table 1 and Figure 3.

	Fluidised bed	Entrained flow
Coal particle size	0.5 mm – 5 mm	0 – 0.5 mm
Coal moisture	Dry	Dry/slurry
Coal type	Non-caking coals	Low- ash coals
Ash in coal	< 60%	< 30%
Gasification agents	Air, steam, oxygen	Oxygen and steam
Temperature	850°C – 950°C	1300°C – 1450°C
Pressure	< 25 bar	< 30 bar
Residence time	0.5 – 1.5 h	< 10 s
Carbon efficiency	65% - 85%	75% - 90%
Gasification efficiency	55% - 75%	55% - 70%
Commercial examples	Winkler	Texaco, Prenflo, Shell & Koppers –Totzek

Table 1. Comparison of fluidised bed and entrained- flow fine coal gasifiers.

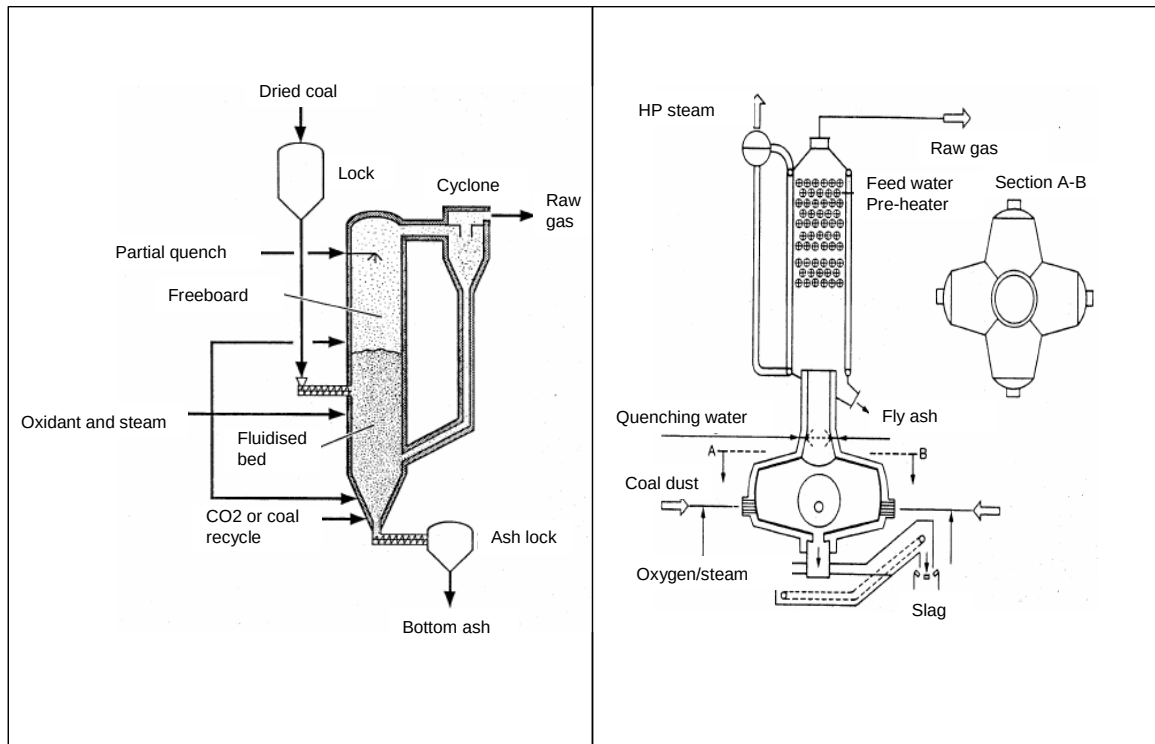


Figure 3. Fluidised bed and entrained flow gasifiers.

The only commercial example of fine coal gasification in South Africa is the six Koppers-Totzek gasifiers which were operated from 1975 to 1999 by African Explosives and Chemical Industries (AECI) at Modderfontein for ammonia production. Gas production was $\pm 100\,000\text{ Nm}^3/\text{h}$ containing 60% CO. The fixed carbon conversion was between 70 and 80% and the gasification efficiency was between 60 and 70%. A pilot fluidised bed gasifier supplied by Krupp Engineering was operated by Highveld Steel and Vanadium Corporation in 1988. The objective of the project was to demonstrate fluidised bed gasification technology for the gasification of the discard coal produced by the surrounding mines in the Witbank area. Problems experienced by the Krupp gasifier included low carbon conversion and clinkering of the coal at the oxygen and steam nozzles in the gasifier.

A better understanding of the gasification characteristics of South African coal is important for fine coal gasifier selection and development.

4 Coal selection and characterisation

A suite of four South African coals has been identified as being possible fuels for power stations which would operate for three or four decades, towards the middle of this century. The selected coals are currently used as fuel for the Lethabo (New Vaal coal), Matla, Matimba (Grootegeeluk coal) and Duvha power stations and are typical of South African power station feed coal.

The coal characterisation tests done on the four selected coals are:

- Proximate analysis (Advanced coal technologies)
- Ultimate analysis (Advanced coal technologies)
- Petrographic analysis (SA Petrographics)
- Thermogravimetric analysis (University of Pretoria)
- BET surface area by N_2 adsorption (Protechnik laboratories)

- FSI and Roga index (Advanced coal technologies)

The laboratories that carried out the analysis are given in brackets. A summary of the analyses is given in Table 2.

	New Vaal	Matla	Grootegeeluk	Duvha
Calorific value (MJ/kg)	15.1	18.6	19.8	21.06
Ash content (%)	40.4	33.4	34.9	32.5
Carbon (%) (maf)*	79.10	80.2	81.8	89.3
Vitrinite reflectance (%)	0.53	0.64	0.68	0.75
Surface Area (m ² /g) ⁺	7.01	2.08	<1	< 1
Porosity (%)	3.2	1.3	1.5	1.1
Reactivity index (hr ⁻¹)	3.02	1.68	1.51	0.92
Free swelling index (FSI)	0	0	1	0
Roga index (RI)	0	0	10	0

Table 2. Coal characterisation parameters.

* Moisture and ash free

⁺ Particle size 1 mm

Due to the high ash content and low calorific value of the selected coals they are classified as low grade (D). The moisture and ash free (maf) carbon content and the vitrinite random reflectance are good indicators of coal rank. Table 2 shows that the selected coals are bituminous in rank since the carbon contents (maf) are between 75% and 85% and the vitrinite reflectance values are between 0.45 and 1.25. New Vaal coal has the lowest rank parameter being closer to the sub-bituminous coals and Duvha has the highest rank parameter being closer to the semi-anthracite coals. Approximately ninety percent of South African coals fall within the rank parameters given above.

The BET analysis measures the surface area and porosity of the coal by means of nitrogen adsorption. Due to more extensive coalification, the older higher rank parameter coals have lower surface areas and porosities as reflected in Table 2.

Thermogravimetric analysis (TGA) is used to measure the reactivity index of coal. Carbon dioxide (CO₂) is reacted with a coal sample at 1000°C and the weight loss is measured as a function of time. The reactivity index (Rs) is given by $R_s =$

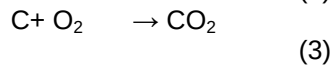
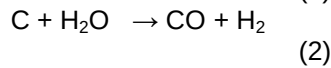
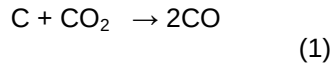
$0.5/t_{0.5}$ with $t_{0.5}$ being the time required for 50% conversion of the fixed carbon. Table 2 shows that the reactivity index increases with increasing surface area and porosity. During the above reactivity test, CO₂ diffuses into the coal, adsorbs on the active sites within the coal and reacts by means of a surface reaction. The rate of the CO₂ gasification reaction would therefore be promoted by higher coal porosity and surface area resulting in a higher reactivity index. The reactivity index is also dependant on the surface chemistry and the catalytic effect of ash in the coal.

The Free Swelling Index (FSI) and Roga Index (RI) analyses are used to measure the caking and agglomerating nature (tendency to deform and stick together) of coal. The FSI is measured on a scale of 0 to 9 with 0 being the least swelling in nature and the RI is measured on a scale of 0 to 90 with 0 being the least sticky in nature. If the coals have caking and agglomerating properties this could potentially be problematic for fluidised bed operation since the coal particles will stick together, de-fluidised and clinkers will be formed in the bed possibly causing defluidisation. Of the selected coals only the Grootegeeluk coal is expected to be

weakly caking or agglomerating in nature as shown in Table 2.

5 Gasification kinetics

The gasification reactions (1 and 2) occur at a much lower rate (up to 1000 times slower) than the combustion reaction (3):



The rate of the gasification reactions therefore has a major effect on the carbon conversion efficiency that can be achieved in a fluidised bed gasifier which operates at moderate temperatures (< 1000°C).

The rate of char conversion is often expressed using the rate equation given below [6, 7].

$$\frac{dX}{dt} = K (1 - X)^\beta P_{\text{CO}_2}^\alpha \quad (4)$$

X - Char conversion (-)
 t - Time (min)
 P_{CO_2} - Pressure of CO_2 (kPa)
 β - Reaction order with respect to the solid phase

α - Reaction order with respect to the gas phase

$$K = k_0 \exp\left(\frac{-E}{RT}\right) \quad (5)$$

k_0 - Pre-exponential factor (min^{-1})
 E - Arrhenius activation energy (J/mol)
 R - Universal gas constant = 8.314 (J/mol.K)
 T - Temperature (K)

The above rate equation is referred to as the grain model and is used to describe the effect of char conversion, CO_2 concentration and temperature on the gasification rate (dX/dt). Equation 4 shows that, as char conversion proceeds initially from $X = 0$ to $X = 1$, the gasification rate slows down. Increasing the temperature and CO_2 partial pressure speeds up the reaction rate.

Thermogravimetric analysis (TGA) is used to determine the parameters in the above model. A Bergbau – Forschung TGA at North West University was used to measure the kinetic parameters for each of the selected coals. The TGA experiments measure the weight loss against time of a pre-weighed sample. The data is converted to a conversion against time graph as shown in Figure 4 for Matla coal.

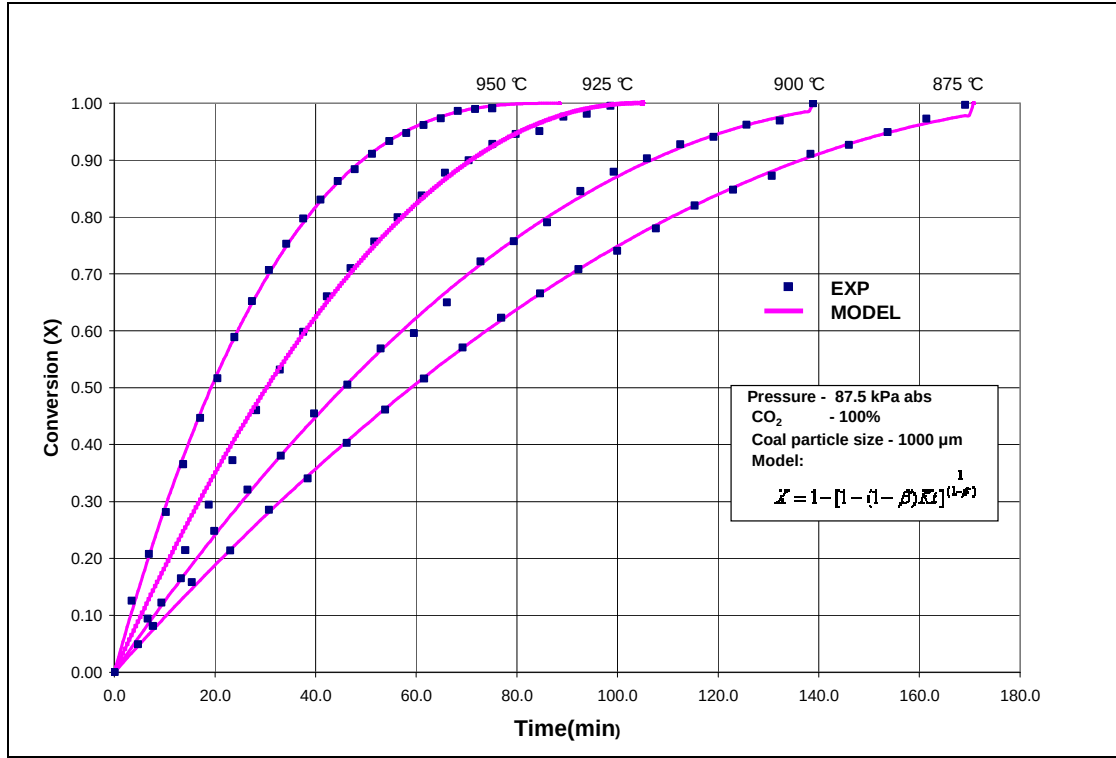


Figure 4. Conversion vs. time graph for Matla coal at four temperatures.

For the TGA tests the CO₂ pressure was maintained at 87.5 kPa. To model the experimental data equation 4 therefore reduces to:

$$\frac{dX}{dt} = K (1 - X)^\beta \quad (6)$$

Integration of equation 6 gives:

$$1 - X = [1 - (1 - \beta)Kt]^{\frac{1}{(1-\beta)}} \quad (7)$$

The model parameters β and K were fitted to the experimental data using least squares regression and are given in Table 3. Figure 4 shows that the model gives a satisfactory fit to the experimental data within the temperature range studied.

Char	Temperature (°C)							
	875		900		925		950	
	k ¹ min ⁻¹	β ²	k min ⁻¹	β	k min ⁻¹	β	k min ⁻¹	B
New Vaal	0.0571	0.69	0.0773	0.62	0.1147	0.62	0.1810	0.63
Matla	0.0098	0.47	0.0127	0.48	0.0194	0.49	0.0316	0.63
Grootegeeluk	0.0031	0.91	0.0061	0.93	0.0090	0.92	0.0166	0.88
Duvha	0.0028	0.80	0.0043	0.71	0.0073	0.83	0.0116	0.87

Table 3. Grain model parameters K and β for selected chars.

The Arrhenius equation can be used to describe the effect of temperature on the reaction rate constant. The Arrhenius equation can also be written as:

$$\ln(K) = \left(\frac{-E}{R}\right)\left(\frac{1}{T}\right) + \ln(k_0) \quad (8)$$

The parameters k_0 (y-intercept) and $-E/R$ (gradient) are obtained by plotting $\ln(K)$

against $1/T$ as shown in Figure 5.

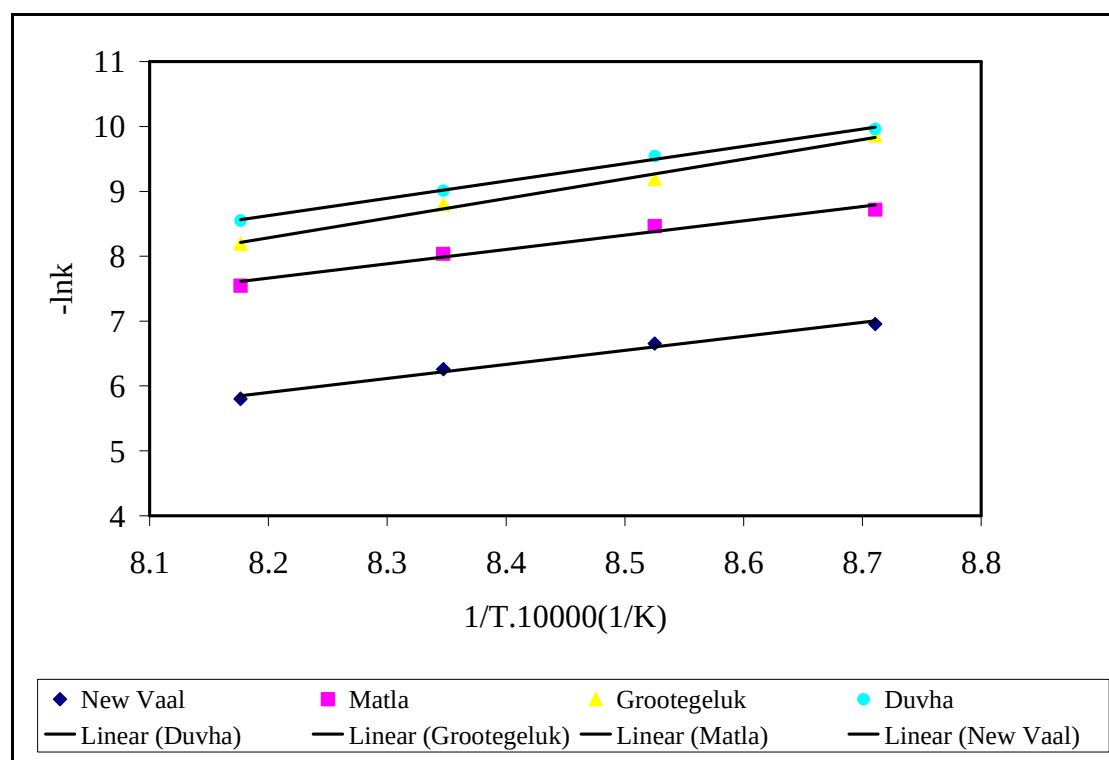


Figure 5. Arrhenius plots for selected coals

The Arrhenius constants and pre-exponential factors are given in Table 4.

Char	E (kJ/mol)	ln(k_0) (s^{-1})	Correlation Coefficient (R^2)
New Vaal	180 (+14)	11.82 (+1.40)	0.9884
Matla	184 (+20)	10.44 (+2.04)	0.9765
Grootegeluk	252 (+16)	16.54 (+1.63)	0.9919
Duvha	222 (+9)	13.29 (+0.88)	0.9970

Table 4. Arrhenius constants

Two special cases of the grain model are the shrinking core model ($\beta = 2/3$) and the homogenous model ($\beta = 1$) (also referred to as the volumetric model). The shrinking core model assumes that the reaction occurs at the external surface of the particle and gradually moves to the middle of the particle leaving an ash layer behind. The homogenous model assumes that the reaction takes place uniformly throughout the whole volume of the particle. The actual reaction normally takes

place simultaneously via both of the above models. A lower value of β indicates that the shrinking core model is the dominant mechanism [6, 7]. The data in Table 3 suggests that, for the Matla coal, the shrinking core model is the dominant mechanism and for the New Vaal coal the homogenous model is dominant.

6 Pilot scale gasification tests

Pilot scale gasification tests were carried out on the four selected coals using an atmospheric pressure air-blown bubbling fluidised bed gasifier.

A flow diagram and specifications of the CSIR pilot fluidised bed gasifier are given in Figure 6 and Table 5.

Features of the pilot plant include:

- A screw feeder is used to feed coal into the gasifier above the bed
- A water-cooled screw is used to remove coarse ash from the bed
- A cyclone is used to removed fly ash from the gas

- The gasification air is preheated by means of a gas-to-gas heat-exchanger
- The area of the freeboard is 7.5 times the area of the bed to increase freeboard residence time
- Online gas analysers are used for gas analysis

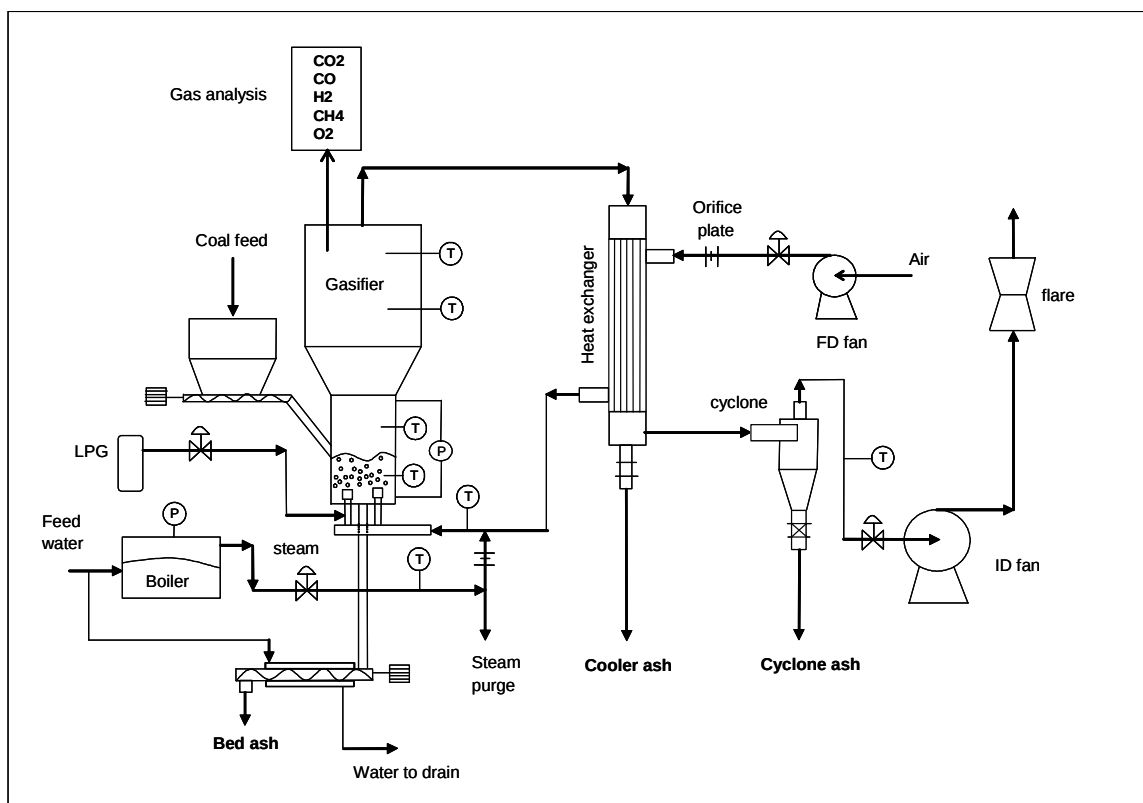


Figure 6. Flow diagram of the pilot fluidised bed gasifier.

Bed dimensions (m)	0.2 × 0.2
Bed area (m ²)	0.04
Freeboard dimensions (m)	0.55 × 0.55
Freeboard area (m ²)	0.3025
Furnace height (m)	4 (2m bed & 2m freeboard)
Fluidised bed height (m)	< 0.6
Coal feed rate (kg/h)	20 to 30
Coal particle size (mm)	< 5
Coal CV (MJ/kg)	15 to 25
Air flow rate (Nm ³ /h)	40 to 60
Steam flow rate (kg/h)	7 to 13
Bed temperature (°C)	900 to 950
Air temperature (°C)	160 to 210
Fluidising velocity (m/s)	1.5 to 2.5
Gas CV (MJ/Nm ³)	2.5 to 3.5
Pressure	Atmospheric
Operating mode	Combustion and gasification
Gas cleaning	Cyclone

Table 5. Specifications of the CSIR pilot fluidised bed gasifier.

Results of the pilot plant tests at 925°C and 950°C are given in Tables 6 .

	New Vaal		Matla		Grootegeeluk		Duvha	
Test number	1	2	3	4	5	6	7	8
Coal feedrate (kg/h)	28.7	23.9	27.0	24.3	23.0	23.0	26.4	26.4
Airflow (Nm ³ /h)	52.2	47.0	50.6	50.9	48.5	47.8	47.5	47.8
Steam flow (kg/h)	9.1	5.8	8.5	8.5	10.2	10.0	10.9	9.0
Air and steam temp. (°C)	202	159	190	185	173	178	176	186
Oxygen: carbon molar ratio	0.48	0.52	0.42	0.47	0.46	0.45	0.34	0.35
Steam: carbon molar ratio	0.50	0.38	0.41	0.46	0.57	0.56	0.47	0.39
Coal particle size (mm) ¹	2.4	1.2	1.6	1.6	1.9	1.9	1.6	1.6
Fluidising velocity (m/s)	2.2	1.9	2.1	2.2	2.1	2.1	2.1	2.1
Mid-bed temperature (°C)	922	947	925	949	927	953	927	949
Lower bed temperature (°C)	967	948	995	972	948	978	921	954
FBG exit temperature (°C)	750	720	752	756	742	764	761	773
Dry gas composition								
CO (%)	NR ²	11.1	10.8	11.6	8.7	10.2	8.8	9.9
H ₂ (%)	NR	8.6	10.0	9.6	9.4	9.5	8.5	9.3
CH ₄ (%)	NR	0.7	0.8	0.7	1.1	1.1	0.8	0.7
CO ₂ (%)	NR	15.8	14.8	14.6	15.0	14.9	15.3	15.0
N ₂ + others ³ (%) ⁴	NR	63.7	63.5	63.4	65.7	64.2	66.5	65
O ₂ (%)	NR	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Gas calorific value (MJ/Nm ³)	-	2.8	3.0	3.0	2.7	2.9	2.5	2.7
Bed pressure drop (Pa)	2664	2115	2553	2259	2553	2553	2455	2456
Char residence time (min)	36.7	36.6	37.4	37.6	45.1	45.1	35.7	35.7
Carbon in bed char (%)	2.8	1.4	24.3	20.8	26.8	26.4	38.6	33.9
Bed char particle size (mm)	1.1	0.6	1.2	1.0	0.8	1.0	0.9	1.2
Carbon in cyclone char (%)	19.5	15.5	32.3	27.8	31.0	27.0	41.6	43.2
Cycl. char particle size (mm)	0.07	0.08	0.05	0.07	0.07	0.07	0.06	0.08
Char elutriated (%)	60.6	66.6	53.8	55.6	51.1	51.6	58.3	59.9
Fixed carbon conversion (%)	82.7	85.9	68.2	74.0	63.2	67.0	52.0	53.7

Table 6. Summary of fluidised bed gasification tests at 90 kPa absolute pressure

¹ d₅₀ – 50% of the coal mass is less than the d₅₀ size

² NR - no reading

³ Others are < 0.4% and include H₂S, NH₃, HCN and C₂⁺

⁴ (N₂ + others) by difference

The results show that:

- A gas calorific value of between 2.6 MJ/Nm³ and 3.0 MJ/Nm³ was obtained
- The gasification efficiency varied between 37% and 48%
- The fixed carbon conversion varied between 52.4% and 82.7%
- Increasing the temperature from 925°C to 950°C improved the carbon conversion by 3.6% on average.
- The carbon conversion increased with decrease in the rank parameter (vitrinite reflectance) as shown in Figure 7.

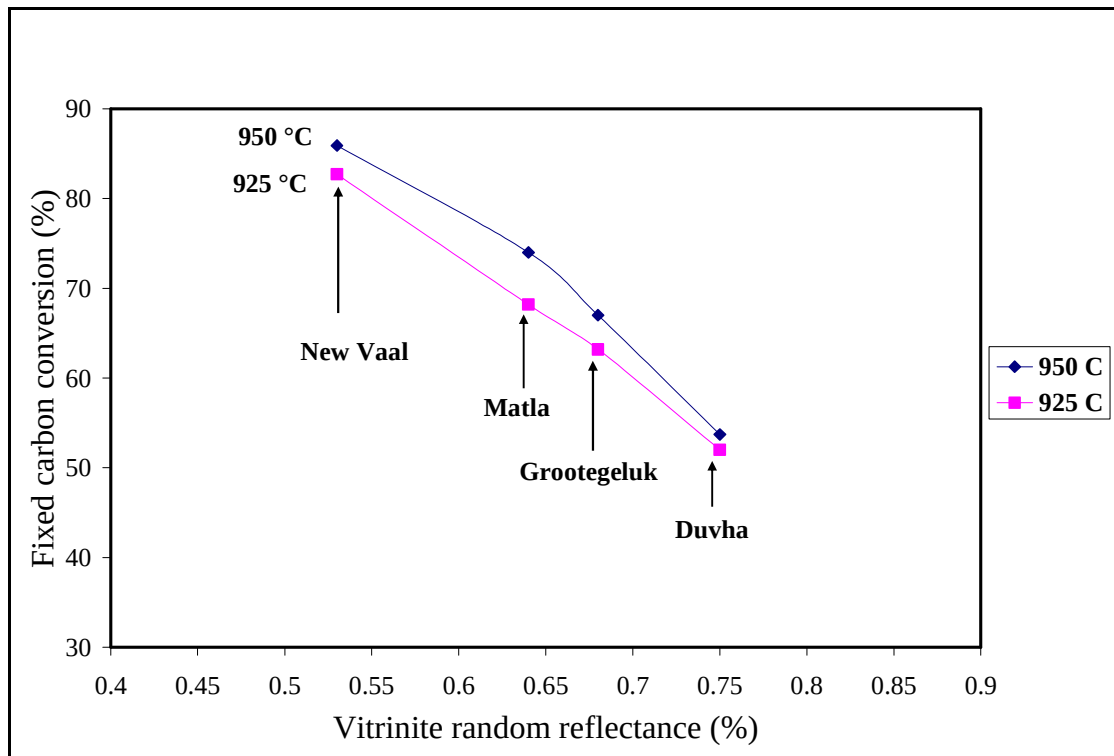


Figure 7. Relationship between fixed carbon conversion and vitrinite random reflectance at 925°C and 950°C.

Other investigators [8] have also found that carbon conversion decreases with increase in coal rank.

The carbon conversion and gas quality of a large-scale fluidised bed gasifier can be improved by:

- Increasing the height of the furnace
- Increasing the freeboard temperature (reduce heat losses)
- Increasing the fluidised bed height
- Increasing the air and steam preheat temperatures.

During the tests it was observed that the temperature above the distributor (30 mm above) was higher (up to 100°C) than the temperature in the middle of the bed. This is due to the partial combustion reaction ($C + O_2 \rightarrow CO + CO_2$) occurring in the bottom 10% of the bed. The average bed temperature therefore has to be maintained well below the ash fusion temperature of the coal to prevent clinkering and agglomeration of the bed

7 Conclusions and recommendations

- Electricity demand in South Africa is increasing at a rate of 1000 MW per year
- Coal generation is likely to provide 80 – 90% of the increased capacity
- New coal power stations need to incorporate clean coal technology (CCT)
- IGCC is a potential CCT that can be applied in South Africa
- Fine coal gasification is a key enabling technology for an IGCC power station
- Fluidised bed gasifiers can utilise high-ash South African coals and therefore are a potential candidate technology for IGCC power stations
- The vitrinite random reflectance of bituminous coal is a good indicator of the carbon conversion that can

be achieved in a fluidised bed gasifier

- Due to the relatively low reactivity of South African bituminous coals a secondary combustion stage may be required after the fluidised bed gasifier to convert the residual carbon in ash.

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